

This question paper contains 4 printed pages]

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S. No. of Question Paper : 2252

Unique Paper Code : 32351403

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Name of the Paper : Ring Theory and Linear Algebra-I

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any two parts from each question.

All questions are compulsory.

1. (a) Describe all the subrings of the ring of integers. $6\frac{1}{2}$
- (b) Let m and n be positive integers and let k be least common multiple of m and n . Show that $m\mathbb{Z} \cap n\mathbb{Z} = k\mathbb{Z}$. $6\frac{1}{2}$
- (c) Show that : $\mathbb{Z}_3[x]/\langle x^2 + x + 1 \rangle$ is not a field. $6\frac{1}{2}$

P.T.O.

2. (a) Find all idempotent and nilpotent elements in $\mathbf{Z}_3 \oplus \mathbf{Z}_6$.
6
- (b) Show that any finite field has order p^n , where p is a prime.
6
- (c) Let R be commutative ring with unity, and let A be an ideal of R . Show that $\frac{R}{A}$ is an integral domain if and only if A is prime.
6
3. (a) Let ϕ be an isomorphism from a ring R onto a ring S . Show that ϕ^{-1} is an isomorphism from S onto R .
6½
- (b) Let R be a ring with unity e and let characteristic of R be n . If $n > 0$, show that R contains a subring isomorphic to $\mathbf{Z}_n$ and if $n = 0$, show that R contains a subring isomorphic to $\mathbf{Z}$.
6½
- (c) Determine all ring homomorphisms from $\mathbf{Z}$ to $\mathbf{Z}$.
6½
4. (a) Prove that a subset W of a vector space V is a subspace of V if and only if $0 \in W$ and $ax + y \in W$ whenever $a \in F$ and $x, y \in W$.
6

- (b) Whether $-x^3 + 2x^2 + 3x + 3 \in \text{span}(\{x^3 + x^2 + x + 1, x^2 + x + 1, x + 1\})$. Justify. 6
- (c) Let V be a vector space over a field of characteristic not equal to two and u, v and w be distinct vectors in V . Prove that $\{u, v, w\}$ is linearly independent if and only if $\{u + v, u + w, v + w\}$ is linearly independent. 6
5. (a) Let V be a vector space having a finite basis. Then show that every basis for V contains the same number of elements. $6\frac{1}{2}$
- (b) Check whether $\{x^2 + 3x - 2, 2x^2 + 5x - 3, -x^2 - 4x + 4\}$ is a basis for $P_2(\mathbf{R})$? $6\frac{1}{2}$
- (c) Let V and W be vector spaces and let $T : V \rightarrow W$ be linear. If V is finite-dimensional, then prove that *nullity* $(T) + \text{rank } (T) = \dim V$. $6\frac{1}{2}$
6. (a) Let β and γ be standard ordered bases for $\mathbf{R}^3$ and $\mathbf{R}^2$, respectively and let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be a linear transformation given by 6
- $T(a_1, a_2, a_3) = (2a_1 + 3a_2 - a_3, a_1 + a_3)$. Compute $[T]_{\beta\gamma}^T$.

- (b) Let V and W be finite-dimensional vector spaces with ordered bases β and γ , respectively. Let $T : V \rightarrow W$ be linear. Then T is invertible if and only if $[T]_{\beta}^{\gamma}$ is invertible. Furthermore, $[T^{-1}]_{\gamma}^{\beta} = ([T]_{\beta}^{\gamma})^{-1}$. 6

- (c) Let $A = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$ and $\beta = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$ be an ordered basis of $\mathbb{R}^2$. Find $[L_A]_{\beta}$ and also find an invertible matrix Q such that $[L_A]_{\beta} = Q^{-1} A Q$, where L_A is a left-multiplication transformation. 6